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The Relevance of Mating Control for Successful Implementation of Honey Bee Breeding Programs

Aleksandar Uzunov , Evert W. Brascamp , Manuel Du  and Ralph B  chler 

This article mainly focuses on the challenges associated with implementing mating control and is the next in line of the series on the concepts of honey bee breeding programs that puzzle many beekeepers around the globe (Uzunov et al., 2017, 2022).

A few years after initiating a breeding program, the honey bee breeders and beekeepers mastered the performance testing and data collection and, with help from a breeding expert, have learned how to select queens based on the list of their estimated breeding values. But, of course, that experience comes with practice, training, and access to tailored guidelines (B  chler et al., 2013; Uzunov et al., 2015). Nevertheless, mating control appears the most complex step towards completing breeding (Uzunov et al., 2017). Unfortunately, when initiating or rebooting breeding programs during the SMARTBEES-project for many honey bee populations around Europe, this challenge of mating control was installed quite late or was still missing even after a few generations of performance testing. Examples of such circumstances are the breeding programs in Malta (*Apis mellifera ruttneri*), the Basque Country in Spain (*A. m. iberiensis*), Macedonia (*A. m. macedonica*, Figure 1), Greece (*A. m. macedonica*, *A. m. cecropia*, and *A. m. adami*), and other regions where mating control, due to a lack of expertise, financial and/or logistic reasons, was still in the “primordial stage”. In contrast, mating control, employing mating stations, and instrumental insemination has been well-known and a long-standing practice in honey bee breeding programs in Germany, Austria, Switzerland, Poland, and the USA (Cobey et al., 2013).

Why is mating control a challenge for many breeding networks and groups? Primarily, mating stations need to be isolated from

other colonies, and suitable places are often scarce due to high colony density and lack of geographic barriers (areas of open water, mountain ridges and areas unpopulated with honey bees or unsuitable for beekeeping). Furthermore, in mountainous regions, accessing those places might be a real logistical or even financial challenge for the beekeepers, which may influence their decision to establish a mating station. On top of that, the costs of establishing and running mating stations are considerable. According to the EurBeST project report, the costs of mating control are close to 10% of the production price of a single queen, which ranges from 12 to 45 EUR/queen in different European countries (European Commission, 2021). Additionally, when it comes to establishing other means of mating control, particularly instrumental insemination, the overall costs may escalate and become a legitimate obstacle to having functional honey bee mating control. Finally, local expertise and support by experienced breeders and bee extensionists play a crucial role in solving the challenge of mating control and enhancing the understanding among the beekeepers of this critical element of breeding programs.

Why controlled mating?

For most beekeepers, open mating is standard practice. Virgin queens leave the hive for one or few mating (nuptial) flights to mate with 10 to 20 random drones from other hives in the broad vicinity (Tarpy & Nielsen, 2002). In the case of a breeding program, but also within common beekeeping practice, this implies that only the selection of the maternal side is controlled by the breeder’s decision from which colonies to raise virgin queens. For the paternal side, the genetic quality of workers in the colony and later queens raised from the colony depends on the quality of drones in a large area around the hive. Of course, one half of the

genetic makeup of the worker bees is derived from the queen and the other half from the drones the queen mated with. The same holds for the expected performance of a colony. In the case of open mating, the breeder, therefore, has no control of the paternal contribution to the expected performance of the colony.

This can be illustrated by the extreme case where the breeder colonies are surrounded by colonies of other beekeepers that do not participate in a breeding program. Let us assume that the genetic quality of the colonies of those beekeepers remains constant (100 units) over generations. Figure 2 illustrates what happens if, for every generation, the breeder selects colonies that are 4 units above his colony average.

In the first generation, the expected genetic level equals the average of that of the selected colonies (104) and the drones the queens raised from the selected colonies are mated to (100). The result is 102. In generation 2, the expected genetic level again equals the average of that of the selected colonies ($102 + 4 = 106$ in generation 1) and that of the drones (100). The result is 103, and so on and so forth. It can be seen that over generations, the genetic level of the breeder’s colonies stepwise approaches to a limit of 104 units (Plate et al., 2019). Compare this with the situation that not only the selected colonies in generation 0 have a level of 104, but also the selected queens that produce the drones (104) (orange trendline). Then the expected genetic level of colonies already in generation 1 would be 104, and in each generation, the genetic level (and the expected performance) would increase by 4 units. Therefore, the differences in breeding success between open and controlled mating become more and more apparent the longer the breeding programs are carried out.



■ **Figure 1.** Direct observations of drone's nuptial flights (drone producing colonies) for assessing the reliability of the island mating station "Snake Island" in Macedonia.

In practice, the situation is more complicated. If, for example, the breeders in the breeding program are in an area with relatively few other beekeepers, the surrounding drones will also improve appreciably, and both the colonies of the breeders and those of the other beekeepers will improve from contributions of the paternal side. Using computer simulations, Du et al. (2021a) studied the situation where a breeding population uses controlled mating and where there is another - called passive - neighbouring population that profits from the genetic improvement in the selection population. It profits in two ways. On the one hand, due to trade, some of the passive population's queens have a dam in the breeding population and on the other hand, some of the drones that passive queens mate with come from the breeding population. Researchers showed that the annual genetic improvement in the passive population eventually equals that in the breeding population but that the passive population lags behind by a number of years. If, for example, 10% of the passive

queens have a dam in the breeding population and 10% of the drones are from a queen in the breeding population, Du et al. found the time lag to be 20 years. This means that when a certain performance level is reached in the breeding population, the same level is expected to be reached in the passive population 20 years later. If the proportions are 90% instead of 10%, the breeding population has a much greater influence on the passive population. Accordingly, the time lag will be shortened to only two years. Another way to look at it is in terms of differences in genetic levels rather than in time. With a genetic improvement of two units per year, a time lag of 20 years means that the passive population will always be 40 units below the breeding population. With a time lag of two years, this genetic difference only amounts to four units, which underpins the benefit of distributing improved material to a broader group of beekeepers.

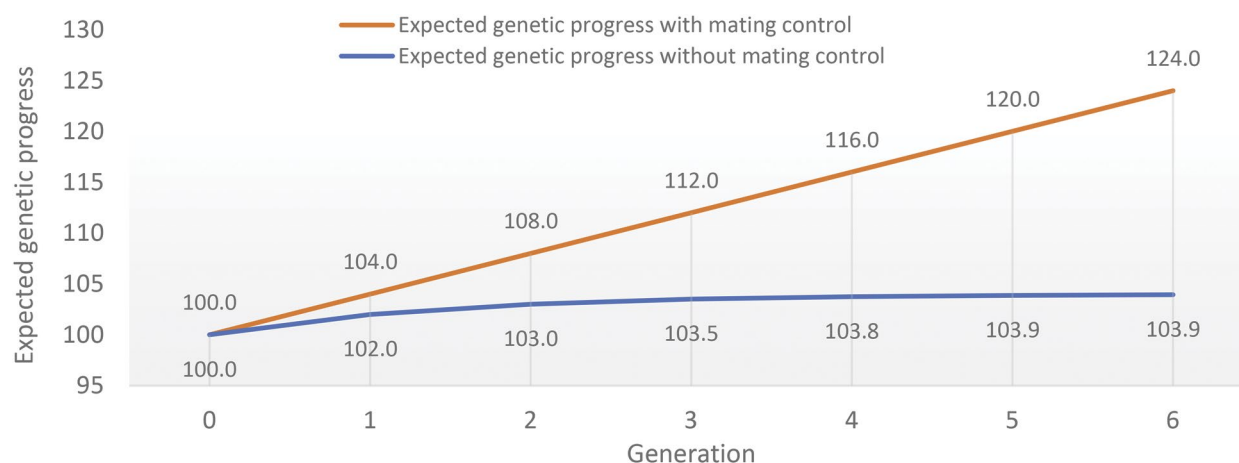
In the arguments for controlled mating that we have brought forward so far, we

made a simplification that the workers are responsible for the colony's performance. However, a colony can only thrive if the queen and worker bees work well together. A queen and her workers interact in diverse ways. By her egg-laying frequency, the queen determines the number of workers, and the workers feed the queen, the queen influences worker behaviour by pheromone release, etc. Consequently, the queen and workers jointly influence most breeding traits (Figure 3). Ideally, a breeding program should strive to improve both the queens' and the workers' genetics. However, this differentiation is often not easy. For example, if a colony produces a lot of honey, this may be caused by the queen producing a strong colony, or by the workers having good foraging skills, or simply by the environmental conditions being favourable.

Plenty of mathematical theory goes into modern breeding value estimation in honey bees that makes it possible to differentiate between the genetic qualities of queens and worker bees and the environment (Brascamp & Bijma, 2014, 2019, Hoppe et al., 2020). However, all of these techniques depend on mating control that works properly. Without it, the genetic parameters and intercorrelation of traits - the significance of which we have explained (Uzunov et al., 2022) - cannot be determined appropriately (Du et al., 2022). Even if correct parameters are at hand, the breeding value estimation will produce biased results (Plate et al., 2019). So, without controlled mating, one not only loses the possibility to select on the paternal side but also the selection of the best queens to produce the next generation will be compromised.

Modes of mating control

When we understand the need and the significance of mating control in honey bee breeding, it is time to itemize what the



■ **Figure 2.** A theoretical model of genetic progress with (orange trendline) and without mating control (blue trendline).

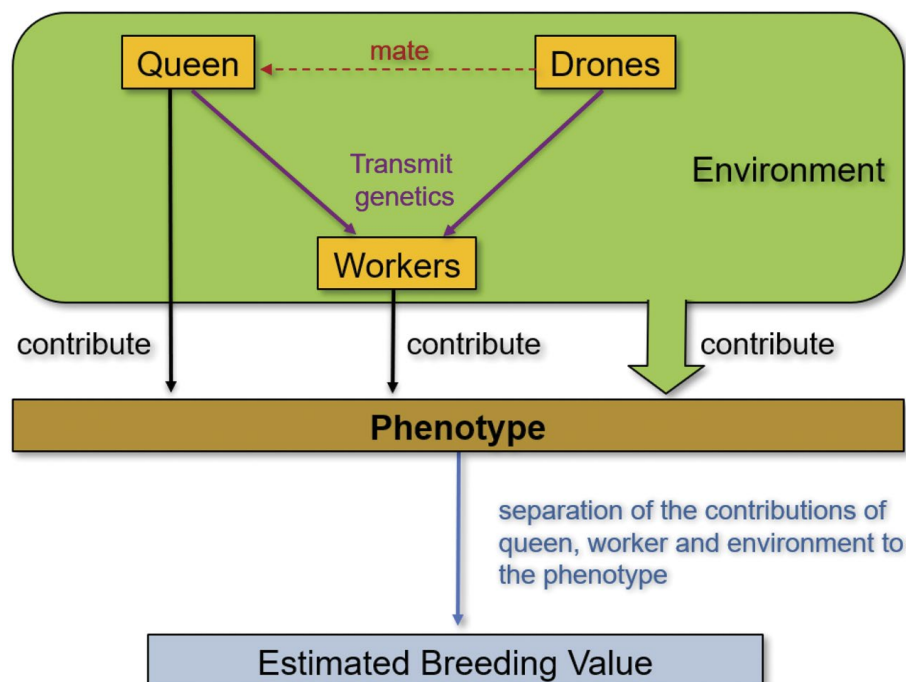


Figure 3. Factors affecting the honey bee colony's phenotype and estimation of breeding values.

beekeepers can do about it. In general, there are four main approaches for achieving mating control in honey bees.

Geographic isolation

Geographic isolation is achieved by any kind of geographic or spatial barrier that restricts the presence of unselected drones in the area where chosen virgin queens fly to mate. For example, a radius of 6 to 10 km ensures controlled mating between selected partners (Büchler et al., 2013; Heidinger et al., 2014; Zakour & Bienefeld, 2014) due to their biological flight limitations; however, based on the orography, such as deep valleys, it could be that a far smaller radius is sufficient. Geographic isolation can be achieved by high mountain ridges, water surfaces for island or peninsula isolation, areas unpopulated with honey bees or areas unsuitable for beekeeping such as deserts or highly industrialised zones. It can further be enforced by legal regulations, which prevent the presence of other colonies. Finding suitable places for such

mating stations can be challenging. Investigation of the location's suitability for setting up a mating station requires the use of customary methods for assessing drone presence and drones and queens' mating behaviour (Büchler et al., 2013; Scheiner et al., 2013).

Instrumental insemination

Instrumental insemination is often limited to research because mating is then controlled most strictly. However, the technique has been used for decades in some European countries, particularly by breeders although rarely in conventional beekeeping operations. Poland is an exception, where it is estimated that in some years, up to 80,000 queens are inseminated (Cobey et al., 2019). This is due to the tradition of having the topic of instrumental insemination in the curricula of many beekeeping schools and extension courses. However, for the majority of the beekeeping world, the technique is still in the hands of the researchers and institutes because it

requires investment in technical and human capacities. In the last decade, semen cryopreservation from populations and genotypes of specific breeding interests has become possible. Besides being a valuable tool for subspecies conservation, in future this technique might be utilised for different breeding initiatives (Viert et al., 2021).

We would strongly emphasise that instrumental insemination ensures the best possibility for control of the mating process when compared to the other means of mating control. This control includes the possibility to mate with drones of one single queen, yielding more accurate breeding-value estimates, and also allows a broader spectrum of drone producing queens used, leading to higher genetic diversity. Note, however, that the extreme case of single-drone insemination may lead to weak colonies that only survive for a few months (De Camargo & Goncalves, 1971). This strategy thus has its main use in experimental designs.

Isolation by saturation - (Biological) isolation

When geographic isolation cannot be achieved, a pragmatic solution is the saturation of the area with drones originating from drone producing colonies (DPC) headed by selected queens. This approach requires understanding, cooperation and coordination among the breeders and the beekeepers in the given area. Commonly, but not exclusively, each breeder is responsible for distributing a sufficient number of selected queens to surrounding beekeepers' colonies which then serve as DPC. In such a way, both the breeder and the beekeepers benefit. The beekeepers have access to the best genetic material, and the breeders ensure the presence of drones from selected stock. Note, however, that still a considerable number of matings will be from unselected stock and that this method is not a good option for the improvement of the breeding population.

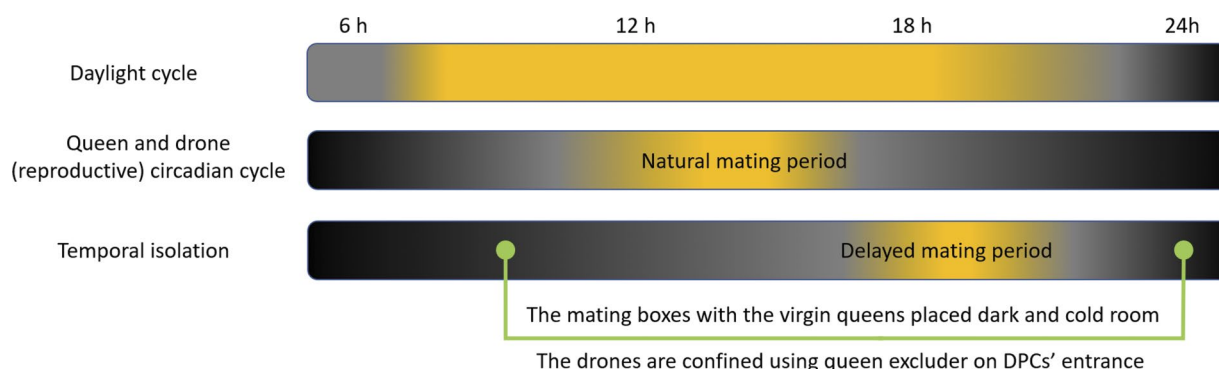


Figure 4. Timeline of conducting temporal isolation during mating season in Central Europe. DPC refers to drone producing colonies.

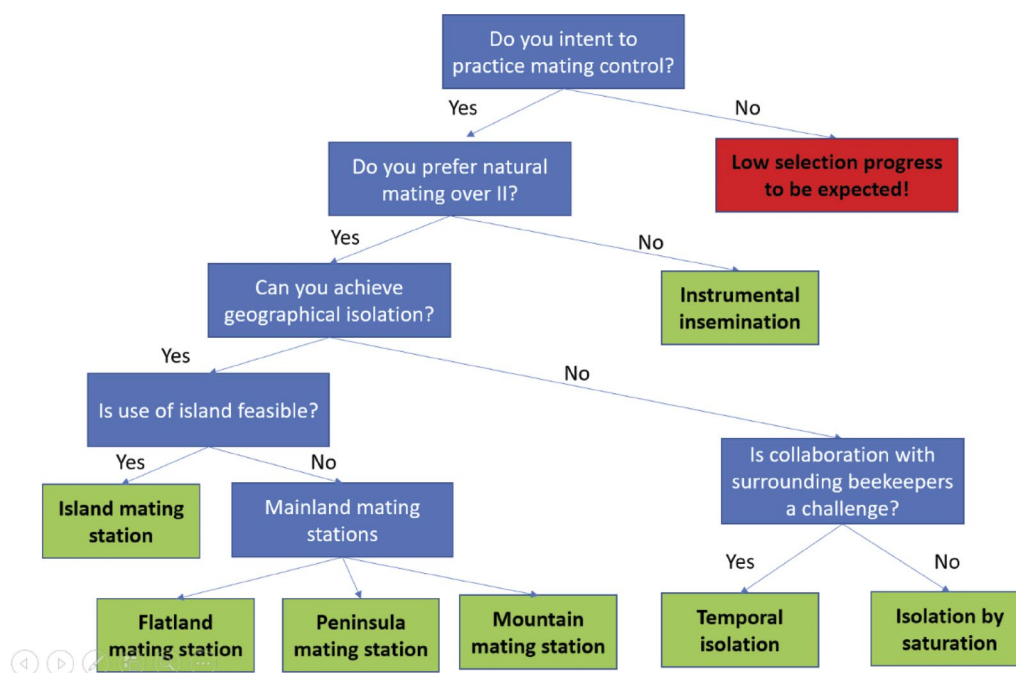


Figure 5. Decision tree for the use of different modes of mating control. Within a breeding program various modes are likely to exist.

Temporal isolation

If isolation is not possible by geographic barriers or saturation, the breeders may achieve controlled mating by deferring the natural mating period. For example, in Central Europe, the mating partners commonly fly to mate from noon to around 5 pm; however, if the mating boxes are contained in a dark and cold room for 22 hours and released after 5 pm, the virgin queens will take a flight stimulated by the light and the ambient temperature (Figure 4). Almost at the same time, the drones are released by removing the queen excluder from the entrance of the DPC's, headed by selected queens. At that time, there is a low probability of having unwanted airborne drones, and there is a good chance of mating the selected partners. Even though it has been known and in limited use for more decades, this approach, with many existing modifications, has been gaining increasing popularity and attention in the last few years (Musin et al., 2021).

Which mode of mating control to choose?

Many factors affect which mode of mating control is most suitable in specific cases. For example, the breeding goal, selection intensity, population structure (subspecies, breed, line), tradition, finances, expertise, logistics, accessibility, orography, colony density, beekeepers' awareness and recognition of breeding, etc. are all critical aspects in deciding what mode is best applied in a particular breeding group. The choice may be different when a breeding program is designed to achieve genetic improvement as compared to a program to propagate improved genetic

material in the broader population. To achieve genetic improvement, it is crucial that pedigrees are correct. That is far less relevant for the propagation of improved material. There it suffices those queens are mated to improved drones. So, to achieve genetic improvement, strict isolation (or instrumental insemination) is to be preferred. For the propagation of improved material, the other modes are suitable alternatives where massive dissemination of the genetic progress into broader population is priority.

A notable example of controlled mating suitable for breeding programs is a network in Germany. In the Beebreed database (www.beebreed.eu) for 2021 there are 81 registered mating stations supplemented by 25 insemination sites. It concerns some 50,000 queens.

Nevertheless, we want to emphasise and demonstrate (Figure 5) that breeders intending to use mating control have various options.

Criticism on mating control

Above, we have given our opinions as to why mating control is crucial for honey bee breeding. However, there are also breeders who strictly reject any form of mating control. An example is the German "Union der Basiszüchter". These breeders argue that the restriction of drones under controlled mating conditions means a narrowing of genetic diversity, while colonies with a diverse range of sire drones generally express

better performance and fitness (e. g., Mattila & Seeley, 2007).

Both these observations are true, yet the consequences for breeding are not as straightforward as they may seem. To fully understand this, one has to go back to the very definition of what breeding is, namely the gradual improvement of a population by selection for heritable traits. The keyword is "heritable", and this is important when applying the results of Mattila and Seeley to breeding activities. Indeed, honeybee colonies with a diverse drone background have a higher genetic diversity of workers and may therefore thrive more than genetically more homogeneous colonies. This higher genetic diversity, however, is not transferred to the next generation as each queen reared from the colony descends from one single drone. The diversity within a next generation colony will depend on the drones the queen mates with but not on the previous generation. The advantages that colonies may have from variability in sire drones can thus not be inherited and is therefore irrelevant for breeding activities. On the contrary, raising a new queen from a colony with diverse sire drones is a gamble because the specific drone that is father to the queen may have good or bad genetic qualities. Under controlled mating, breeding success will be much more consistent.

It should be noted that Du et al. (2021b) showed that the introduction of controlled mating actually yields a shift rather than a reduction of genetic diversity. While the genetic differences within a colony are indeed reduced, the differences between different colonies will be pronounced more

distinctively. In this way, good and bad colonies can be distinguished more easily, and as a result, controlled mating facilitates favourable selection decisions.

Another concern that several breeders have voiced towards the introduction of mating control is that their colonies might suffer from inbreeding depression after some time. Indeed, inbreeding is a phenomenon that will hardly occur under open mating. Nevertheless, within breeding programs, it is a problem that can easily be addressed. For example, the Central European population of *A. m. carnica* has been bred with strong selection under heavy use of mating control, and the inbreeding rates are very acceptable (Hoppe et al., 2020). Furthermore, simulation studies have shown that sustainable breeding with controlled mating is also possible for small populations with only about 200 colonies per year (Plate et al., 2020). In Uzunov et al. (2022), we have given advice on which precautions should be taken so that inbreeding does not reach levels that are out of hand. It is important to monitor the increase in average inbreeding coefficient per generation and to keep that below 1% or even better below 0.5%.

The breeding success achieved in some of the Central European breeding programs, alongside the recent research developments described above, ignited significant interest in incorporating controlled mating in many honey bee breeding programs. Still, a universal solution for overcoming breeders' challenges does not exist since each breeding initiative is characterised by unique circumstances, some of which are out of beekeepers' control. Therefore, the success is obtainable only by joint efforts of the stakeholders that are relevant to implementing breeding programs. Such holistic strategy was chosen from the ongoing BeeConSel project (www.beeconsel.eu). It aims for incorporating various modes of mating control in the breeding programs in three South-Eastern European countries where the scientific expertise, technical assistance, administrative support and cooperation between all mentioned stockholders provides a good chance for achieving success.

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